

A near-zero carbon emission methanol production through CO₂ hydrogenation integrated with renewable hydrogen: Process analysis, modification and evaluation

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ABSTRACT

Carbon dioxide hydrogenation to produce methanol has the potential to liberate humanity from its reliance on fossil fuels, while simultaneously reducing carbon dioxide and developing the economy. This paper focuses on the effect of catalyst type and the process conditions such as temperature, pressure, H₂/CO₂ feed ratio, and space velocity on CO₂ conversion and CH₃OH selectivity. The net generated CO in the reaction system results in the loss of CO₂ and H₂ to the purge gas, which reduces the utilization efficiency of carbon and hydrogen. The specially designed catalysts and processes with recycling can improve CO₂ conversion and CH₃OH selectivity, and the CO flow rate difference can be close to zero in the total recycling condition. Starting from these ideas, the two kinds of near-zero carbon emission processes with total recycling (respectively named total recycling process after flash separation and named total recycling process after stripping separation) are proposed. The utilization efficiency of carbon and hydrogen are similar in the two processes, but hot and cold utilities in the latter are 17.96% and 15.11% lower than in the former, respectively. Through the process integration between reaction heat and double-effect distillation, the energy consumption is further reduced in the total recycling process after stripping separation. The utilization efficiency of CO₂ is 99.88% in the total recycling process after stripping separation with double-effect distillation of the light split/reverse configuration, and the proposed process just requires cold utility.

1. Introduction

Fossil fuels have historically served as the principal energy source for driving global economic growth. With the low-carbon and green transformation of the international energy structure, vigorously developing clean energy, such as renewable energy, is critical to reducing our reliance on fossil fuels (Kumar et al., 2023). Due to the inherent defects of renewable energy such as intermittently, volatility and randomness, it is the current development direction to convert CO₂ emitted from industrial production into stable electro-chemical energy storage medium methanol (Sollai et al., 2023).

Methanol is readily transported and distributed in a liquid form,

enabling it to be utilized in existing energy infrastructures to produce liquid fuel. Methanol has been regarded as “liquid sunshine” due to its use as a chemical energy storage medium for electric with high energy storage density, good stability, and ease of transportation (Trudewind et al., 2014). Methanol also serves as a crucial fundamental chemical material that can be used to produce formaldehyde, acetic acid, olefins, and other compounds (Matzen and Demirel, 2016). CO₂ hydrogenation to methanol with green hydrogen provides a double-effect feasible process route that reduces CO₂ emissions and improves the consumption capacity of renewable energy.

Based on the carbon emission intensity, the hydrogen production process can be classified into gray hydrogen, obtained from coal, blue

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Table 1
Performance comparison of different water electrolysis technologies.

	ALK	PEM	SOEC
Temperature	70–90 °C	70–80 °C	600–1000 °C
Current density	<0.8 A/cm ²	1–4 A/cm ²	0.2–0.4 A/cm ²
Energy consumption	4.5–5.5 kWh/ Nm ³	4.0–5.0 kWh/ Nm ³	2.6–3.6 kWh/ Nm ³
Efficiency	60–75%	70–90%	85–100%
Hydrogen purity	≥99.8%	≥99.99%	/
Industrialization	Yes	Yes	No
Single unit hydrogen production scale	0.5–1000 Nm ³ / h	0.01–500 Nm ³ / h	/
Age of system	10–20 years	10–20 years	/
Advantages	Well industrialized Non-noble catalysts Long-term stability Relative low cost	High current densities Good partial load range Rapid system response High gas purity	Efficiency up 100% Non-noble catalysts High pressure operation
Disadvantages	Low current densities Low partial load range Low dynamics Low pressure	High cost of components Acidic environment Possibly low durability	Laboratory stage Bulky system design Durability No dependable cost

hydrogen, obtained from natural gas, and green hydrogen, obtained from renewable energy sources via electrolytic water. The primary goal of developing the hydrogen industry is to achieve zero carbon or low carbon emissions. Hence, gray and blue hydrogen will be gradually replaced by green hydrogen based on renewable energy, which is the future direction of the energy industry. There are three primary techniques for producing hydrogen from water electrolysis, including alkaline water electrolysis (ALK), proton exchange membrane electrolysis (PEM), and solid oxide electrolysis (SOEC) (Falcão and Pinto, 2020). Among these, PEM electrolysis exhibits greater flexibility in operation and is better suited to the volatility of renewable energy, making it a promising pathway for producing green hydrogen (see in Table 1). Thus,

Table 2
Carbon efficiency of CO₂ hydrogenation to methanol process with recycling.

Research Institution	Catalyst type	Reactor type	Reaction or inlet temperature, °C	Pressure, bar	GHSV, h ⁻¹	Recycle ratio	Carbon efficiency, %	Ref.
Heriot-Watt University	Cu/ZnO/ Al ₂ O ₃	Adiabatic fixed bed	230	50	N.A.	7.9	92.6–93.6	Mignard (2003)
University of Maribor	Cu/ZnO/ZrO ₂	N.A.	N.A.	N.A.	N.A.	N.A.	84.5	Anicic et al. (2014)
Universidade Federal do Paraná	Cu/ZnO/ Al ₂ O ₃	Adiabatic fixed bed	210	75	N.A.	5.0	87.2	Van-Dal and Bouallou (2013)
European Commission, Joint Research Centre, Institute for Energy and Transport	Cu/ZnO/ Al ₂ O ₃	Adiabatic fixed bed	210	76	N.A.	N.A.	88.6	Pérez-Fortes et al. (2016)
Chinese Academy of Sciences	Cu/ZnO/ Al ₂ O ₃	isothermal	250	50	N.A.	N.A.	86.0	Gu et al. (2022)
AkzoNobel-Research	Cu/Zn/Al/Zr	isothermal	250	50	N.A.	5.47	99.83	Kiss et al. (2016)
Hefei University of Technology	Cu/ZnO/ Al ₂ O ₃	isothermal	250	50	N.A.	N.A.	87.5	Yang et al. (2022)
Korea Institute of Science and Technology (KIST)	Cu/ZnO/ Al ₂ O ₃	Adiabatic fixed bed	250	70/80	N.A.	3.6/4.5	66.9–70.5	Joo et al. (2004)
University of Cagliari, Dept. of Mechanical	Cu/ZnO/ Al ₂ O ₃	Adiabatic fixed bed	210	65	N.A.	2.37	95.1	Sollai et al. (2023)
Chemical Process and Energy Resources Institute	Cu/ZnO/ Al ₂ O ₃	Adiabatic fixed bed	250	65	N.A.	3.7	35	Atsonios et al. (2016)
MINES ParisTech, PSL-Research University	Cu/ZnO/ Al ₂ O ₃	Isothermal	260	80	15000	1.5	46	Rivera-Tinoco et al. (2016)
NIRE/RITE	Cu/ZnO/ Al ₂ O ₃ /ZrO ₂ / SiO ₂	isothermal	250	50	10000	N.A.	99.7	Saito et al. (2000)

Note: Carbon efficiency = $\text{CH}_3\text{OH}_{\text{product}}/(\text{CO} + \text{CO}_2)_{\text{feed}}$.

in this study, we have chosen PEM technology for green hydrogen production.

It is important to note that this study does not focus on technical routes for hydrogen production but rather emphasizes the use of green hydrogen. For CO₂ hydrogenation to methanol process, since the chemical inertness and stability of CO₂, an efficient and catalyst is one of the crucial factors to realize CO₂ hydrogenation to methanol. The catalysts have an important effect on the CO₂ conversion rate and CH₃OH selectivity. The catalysts can be generally categorized into three groups based on the various active components. Cu-based catalysts such as Cu/ZnO/Al₂O₃ and Cu/ZnO/ZrO₂ are the most extensively researched catalytic systems (Liu et al., 2003; Pavlišić et al., 2020). The supported precious metal catalysts such as Pd/SiO₂ and Pd/Ga₂O₃ are not suited for large-scale industrial use due to their high price (Jiang et al., 2015; Qu et al., 2014). The oxide catalysts such as ZnO/ZrO₂/In₂O₃ have good internal hydrothermal stability (Martin et al., 2016; Sun et al., 2015; Wang et al., 2017).

However, on account of the thermodynamic equilibrium limit for the CO₂ hydrogenation to methanol, it adversely causes low CO₂ single-pass conversion efficiency. According to experimental results on CO₂ conversion in a fixed-bed reactor with various catalyst types, the conversion rate of CO₂ ranged from 1.5% to 25.9% (Saeidi et al., 2014). The unconverted raw gas must be recycled to increase the utilization rate of raw materials.

As shown in Table 2, there is a significant difference in carbon utilization owing to catalyst type and the presence of purge gas. The purge gas mainly includes CO, CO₂, and H₂. The composition of the purge gas is affected by the reaction temperature and pressure. Some research related to the effect of catalyst type is described in the following subsection. And the present of purge gas affects the utilization efficiency of carbon and hydrogen, further causing controversy on the economics of CO₂ hydrogenation to methanol. Mignard (2003) and Anicic et al. (2014) found that the cost of CO₂-to-methanol production is potentially competitive with fossil-to-methanol production. In comparison, Pérez-Fortes et al. (2016) and Tremel et al. (2015) found the cost of producing hydrogen, which comprises the electrolyzer and power capital costs, is typically the largest component of the overall cost. It is necessary to discuss the economy of CO₂ hydrogenation to methanol under the condition of maximizing carbon and hydrogen efficiency.

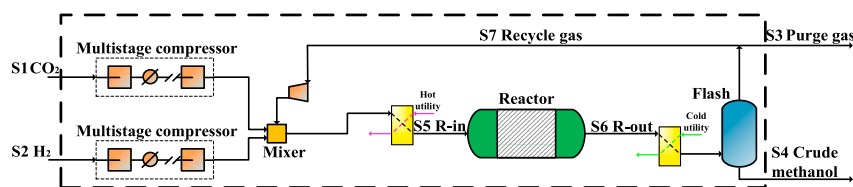


Fig. 1. Typical recycling process of CO₂ hydrogenation to methanol.

Based on the process analysis for reaction unit with recycling, even if the generated CO in the system is small amount, the loss of CO₂ and H₂ cannot be ignored. The key to reduce the CO₂ and H₂ losses is to reduce the difference between inlet and outlet CO flow rate. Another, the CO₂ recovery from crude methanol also is an important issue. To maximize the utilization of CO₂ and H₂, a near-zero carbon emission methanol production through CO₂ hydrogenation integrated with green hydrogen is developed in this study. It is followed by a discussion of the effects of process conditions such as catalyst, temperature, pressure, GHSV (gas hourly space velocity), H₂/CO₂ ratio on CO₂ conversion, methanol selectivity, and purge gas. Through process integration between reaction heat and double-effect distillation, a novel near-zero carbon emission process that does not require external heating utility but just cold utility is introduced. The techno-economic evaluation for the novel process is also conducted.

2. Literature review

There are three primary techniques for producing hydrogen from water electrolysis (Akhtar et al., 2023), including alkaline water electrolysis (ALK), proton exchange membrane electrolysis (PEM), and solid oxide electrolysis (SOEC). Among these, PEM exhibits greater flexibility in operation and is better suited to the volatility of renewable energy, making it a promising pathway for producing green hydrogen (Temiz and Dincer, 2022). Thus, in this study, we have chosen PEM technology for green hydrogen production. ALK technology boasts decades of application experience and is widely regarded as the most developed industrialized method. Conversely, the PEM technology has rapidly gained ground and is now a widespread practice (Shaygan et al., 2019).

The PEM hydrogen production technology boasts a quick start-stop capability and is capable of synchronizing with the fluctuation of renewable energy power generation. As a result, this technology is increasingly becoming the mainstream technology in the production of hydrogen (Ayers et al., 2016). Currently, the demonstration application for PEM hydro-electrolysis hydrogen production has reached the 10 MW level, while the development of 100 MW PEM electrolytic cell is underway. Leading companies in the field of technology and equipment manufacturing, such as NEL-Proton, SIEMENS, and ITM Power, are at the fore. Proton Onsite PEM hydrogen electrolysis devices have been deployed in over 2000 sets, distributed in 72 countries and regions. The company holds 70% of the global PEM hydrogen electrolysis market share and has the ability to integrate hydrogen production systems over 10 MW. Giner's single PEM electrolytic cell has a specification that can generate 5 MW, with a current density that exceeds 3 A/cm². The prototype 50 kW electrolytic cell has completed an accumulated high-voltage operation time of over 150,000 h. PEM hydrogen production technology research is being carried out by Dalian Institute of Chemical Physics, Chinese Academy of Sciences, 718 Institute of China Shipbuilding Industry Group, and other domestic units (Zhao et al., 2018).

For reactor, compared with fixed bed reactors, the studies on slurry bed always show a higher CO₂ conversion rate and milder reaction conditions. Zhang et al. (2016) used Cu/ZnO/Al₂O₃/ZrO₂ catalyst to produce methanol from CO₂, total CO₂ conversion was 25.9% and CH₃OH selectivity was 61.5%. Membrane reactor combines both reaction and separation in the same equipment, when used in CO₂

hydrogenation, it can remove products instantly which is good for CO₂ hydrogenation. Gallucci et al. (2004) studied a zeolite membrane reactor for CO₂ hydrogenation to methanol, the results showed that the one-way conversion of CO₂ was 11.6%. Chen (2004) designed a silicone rubber/ceramic composite membrane reactor, the optimal CO₂ one-way conversion was 6.4%. The membrane reactor has not yet achieved the desired results. A lot of literatures focus on changing the reactor structure, but the downside is that the reactor design is complicated and expensive.

The carbon efficiency upon different catalyst in various reactor for the recycling process are listed in Table 2, which is elaborately described the existing research related to this field and contribution. The following subsections are provided to illustrate the related research and gap in this direction. However, in these literatures, there is no in-depth study on process parameters and the problem of existence of purge gas, nor is there any comparison from the perspective of technology and economy.

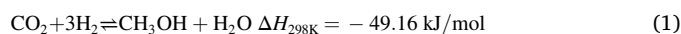
Therefore, this paper selects the most common commercial catalyst and a highly active catalyst as the case, but not limited to these two catalysts, to maximize the utilization of CO₂ from the perspective of process configuration.

3. Analysis of traditional technology

The experimental results of CO₂ conversion in fixed bed reactors with different catalyst types show that the conversion rate of CO₂ is between 1.5% and 25.9%. To improve the utilization rate of raw materials, the unconverted raw gas must be recovered. Due to the catalyst type and the presence of purge gas, the carbon utilization rate is significantly different. The existence of purge gas affects the utilization efficiency of carbon and hydrogen and further causes the economic problem of CO₂ hydrogenation to methanol. Therefore, it is necessary to study the effects of catalyst type and process conditions (such as temperature, pressure, H₂/CO₂ feed ratio, GHSV) on CO₂ conversion and methanol selectivity.

3.1. Effect of catalysts and process parameters

This section focuses on the effect of catalyst type and the process conditions such as temperature, pressure, H₂/CO₂ feed ratio, and GHSV on CO₂ conversion and methanol selectivity. This paper selects the most common commercial catalyst and a highly active catalyst as the case. A highly active Cu/ZnO/Al₂O₃/ZrO₂ fiber catalyst specially developed for CO₂ hydrogenation to methanol (named Catalyst A) (Kiss et al., 2016) and a commercial Cu/ZnO/Al₂O₃ catalyst (named Catalyst B) (Van-Dal and Bouallou, 2013) are selected for comparative analysis. Cu/ZnO/Al₂O₃ catalyst is a commercial catalyst, which is the most widely used catalyst at present, and Cu/ZnO/Al₂O₃/ZrO₂ catalyst is the catalyst that can make the methanol yield reach almost 100% statistically in the chemical process. These two catalysts are selected as representatives from many catalysts to study the influence of catalyst properties and the presence of purge gas in traditional processes and to demonstrate the advantages of the newly designed process flow in this study. The main reactions are shown as following (Atsonios et al., 2016; Dongliang et al., 2021).



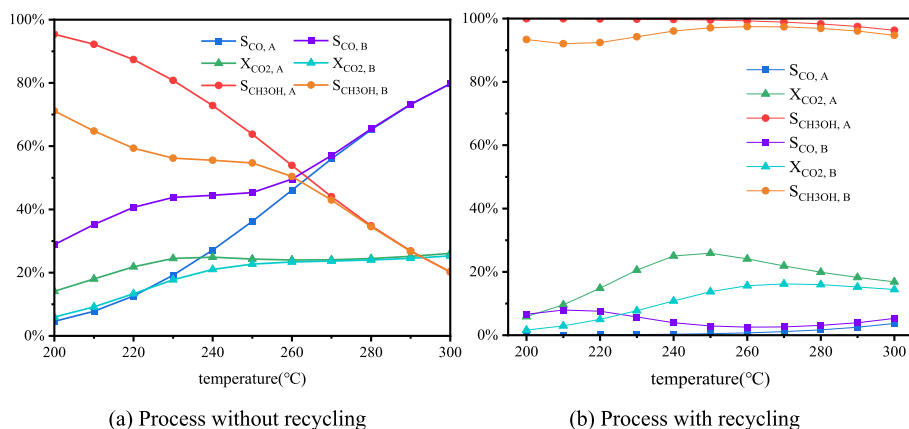


Fig. 2. Effect of temperature on CO₂ conversion and CH₃OH/CO selectivity. (Conditions: $p = 50$ bar, $H_2/CO_2 = 3$, GHSV = 500 h⁻¹, and $R_p = 1\%$).

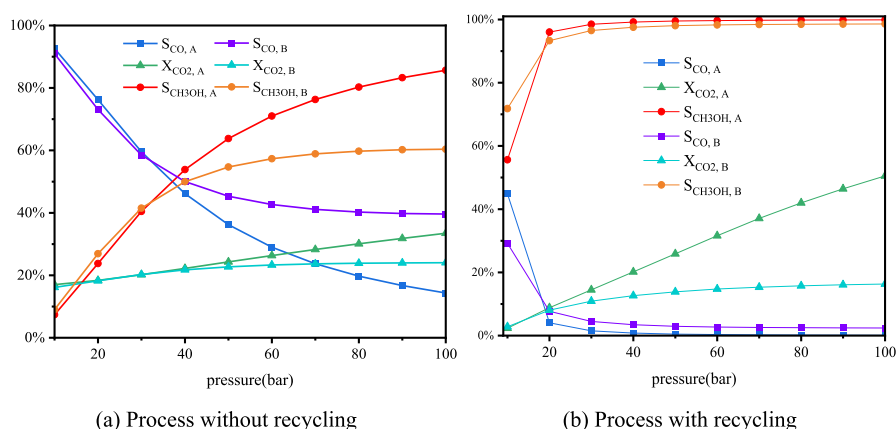


Fig. 3. Effect of pressure on CO₂ conversion and CH₃OH/CO selectivity. (Conditions: $T = 250$ °C, $H_2/CO_2 = 3$, GHSV = 500 h⁻¹, and $R_p = 1\%$).

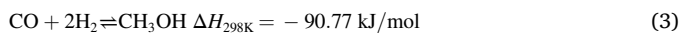
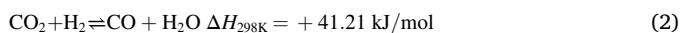


Fig. 1 depicts a typical process of CO₂ hydrogenation to methanol. The reaction product gas is cooled and then sent to a flash separation. The liquid phase from flash unit is the crude methanol for further purification and the vapor phase is divided into purge gas and recycle gas. The recycle gas is pumped and mixed with the fresh feed gas to enhance the raw utilization.

The Rplug model is used to simulate the reaction unit, the reaction kinetics models are describing in supporting information (Tables S1–S4). Process parameters are investigated in the following ranges: $T = 200\text{--}300$ °C, $p = 10\text{--}100$ bar, $H_2/CO_2 = 2\text{--}4$, GHSV = 100–20000 h⁻¹.

The effects of temperature to the reaction unit are presented in Fig. 2. In the absence of recycling, the results characterize the single-pass reaction performance of the catalyst. As can be seen from Fig. 2(a), the CO₂ conversion rate initially increases with the increment of reaction temperature and then remains roughly constant above 250 °C,

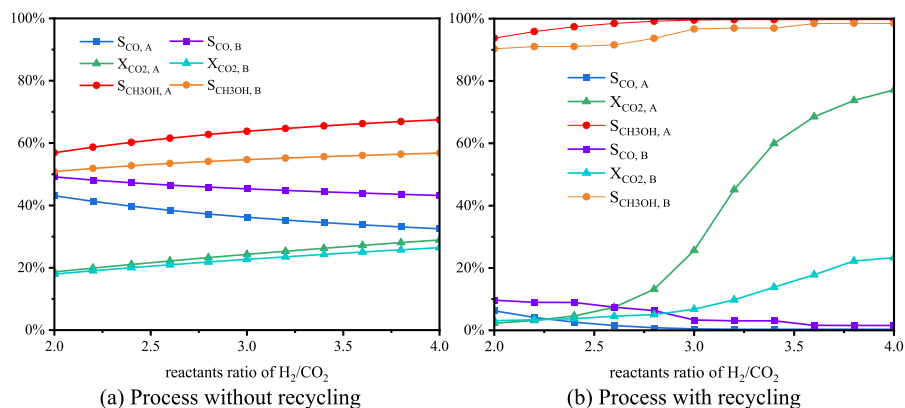


Fig. 4. Effects of H_2/CO_2 on CO₂ conversion and CH₃OH/CO selectivity. (Conditions: $T = 250$ °C, $p = 250$ bar, GHSV = 500 h⁻¹, and $R_p = 1\%$).

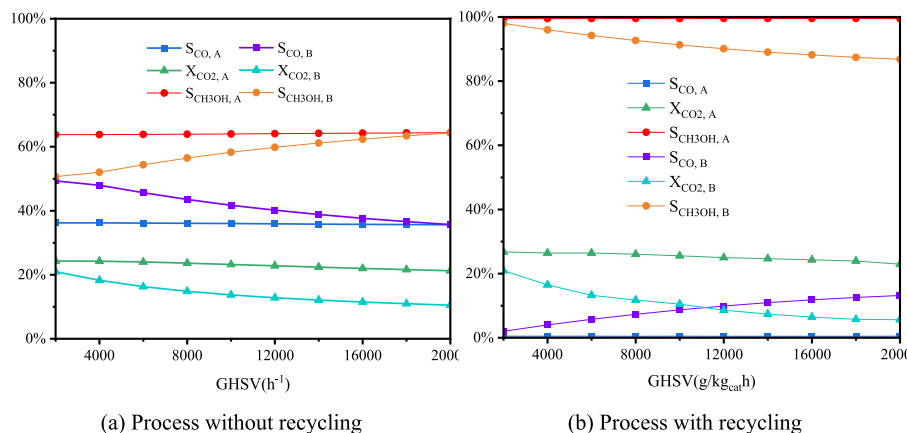


Fig. 5. Effect of GHSV on CO_2 conversion and $\text{CH}_3\text{OH}/\text{CO}$ selectivity. (Conditions: $T = 250^\circ\text{C}$, $p = 250$ bar, $\text{H}_2/\text{CO}_2 = 3$, and $R_p = 1\%$).

indicating that the hydrogenation reaction is kinetically influenced at low temperature and thermodynamic equilibrium limited at high temperature (Basini et al., 2022). The methanol selectivity significantly lessens with the rise of reaction temperature (from 95.40% at 200°C to 20.27% at 300°C for catalyst A, and from 71.15% at 200°C to 20.25% at 300°C for catalyst B). In contrast, the CO selectivity significantly increases. At reaction temperatures below 250°C , there are remarkable differences between the two catalysts in terms of CO_2 conversion and methanol selectivity, while the performance of the two catalysts tends to be consistent when the temperature is higher than 260°C . As shown in Fig. 2(b), the CO_2 conversion in the process with recycling first increases and then slowly lessens as the reaction temperature growing, with a certain peak for each catalyst. The methanol selectivity reduces continuously from 99.81% at 200°C to 96.28% at 300°C for catalyst A, while methanol selectivity remains between 94.23% and 97.47% for catalyst B. In the process conditions with or without recycling, the performance of the two catalysts shows significant differences under the effects of temperature. Both catalysts could maintain a high methanol selectivity in the process with recycling under the influence of temperature, compared with the process without recycling.

The effects of pressure to the reaction unit are presented in Fig. 3. As can be seen in Fig. 3(a) for process without recycling from 10 bar to 100 bar, the CO_2 conversion continuously rises from 17.03% to 33.46% for catalyst A and from 16.11% to 24.03% for catalyst B, the CH_3OH selectivity also rises but the CO selectivity reduces. As shown in Fig. 3(b) for process with recycling from 10 bar to 100 bar, the CO_2 conversion shows an ascending trend, and the catalyst A has the greater trend. The CH_3OH selectivity grows sharply when the pressure increases from 10 bar to 20 bar, and slowly rises when the pressure is higher than 20 bar. The difference in $\text{CH}_3\text{OH}/\text{CO}$ selectivity for the used two catalysts

diminishes in the process with recycling.

Fig. 4 presents the effects of H_2/CO_2 on CO_2 conversion and $\text{CH}_3\text{OH}/\text{CO}$ selectivity. As shown in Fig. 4(a), the CO_2 conversion rate approximately linearly grows with the increment of the feed H_2/CO_2 in the process without recycling, and the CH_3OH selectivity for the two catalysts are slightly improved. The CH_3OH selectivity for catalyst A ranges from 56.90% to 67.50% while that for catalyst B ranges from 50.83% to 56.81%. According to Fig. 4(b) for process with recycling, the CO_2 conversion grows as the reaction feed H_2/CO_2 increases, especially for catalyst A when the feed H_2/CO_2 is greater than 2.6. In terms of selectivity, both catalysts could maintain a high CH_3OH selectivity in the process with recycling, compared with the process without recycling. The CH_3OH selectivity ranges from 93.74% to 99.67% for catalyst A and 90.33%–98.45% for catalyst B.

As shown in Fig. 5, the CO_2 conversion continuously declines with the increase of GHSV, and the effect of GHSV keeps the catalytic reaction activity of catalyst A at a higher level than that of catalyst B. The high space velocity is beneficial to improve the CH_3OH selectivity in the process without recycling, but the improvement is not significant. From Fig. 5(b), under the condition with recycling, the CH_3OH selectivity for catalyst A remained around 99.52% from $\text{GHSV} = 2000$ to 20000 h^{-1} , while the CH_3OH selectivity for catalyst B decreases from 97.99 to 86.82% with the rise of GHSV.

Catalyst A presents better performance than catalyst B in terms of CO_2 conversion and CH_3OH selectivity. Meanwhile, the recycling process can enhance the CH_3OH selectivity, but it is indispensable to investigate the effect of purge gas to avoid the accumulation of generated components in the reaction system. Catalyst A is chosen to discuss the effects of process conditions such as temperature, pressure, GHSV, and H_2/CO_2 ratio on purge gas.

Table 3

Effect of temperature on purge gas. (Conditions: $p = 50$ bar, $\text{H}_2/\text{CO}_2 = 3$, $\text{GHSV} = 500\text{ h}^{-1}$, and $R_p = 1\%$).

T °C	ΔF_{CO} kmol/h	S3				S4		
		CO kmol/h	CO_2 kmol/h	H_2 kmol/h	CH_3OH kmol/h	CO kmol/h	CO_2 kmol/h	CH_3OH kmol/h
200	0.1801	0.1779	5.2475	32.3830	0.1248	0.0026	5.4059	380.0294
210	0.2977	0.2957	6.2540	35.7775	0.1406	0.0039	5.7271	378.6236
220	0.4811	0.4793	7.2866	39.8288	0.1588	0.0056	5.9048	377.2031
230	0.7694	0.7664	8.4609	44.2575	0.1793	0.0079	6.0654	375.5751
240	1.2167	1.2075	9.7524	49.2158	0.2026	0.0110	6.1704	373.7286
250	1.8857	1.8774	11.1664	54.7941	0.2293	0.0151	6.2142	371.6176
260	2.8811	2.8465	12.4121	61.7722	0.2604	0.0199	6.0243	369.3119
270	4.3728	4.3282	14.0531	69.0091	0.2962	0.0263	5.9311	366.2643
280	6.5654	6.4790	15.6418	77.7584	0.3388	0.0338	5.6776	362.6018
290	9.7003	9.6271	17.3224	88.0232	0.3899	0.0426	5.3411	358.0905
300	14.2451	14.1325	18.9015	100.6119	0.4521	0.0520	4.8663	352.3405

Table 4Effect of pressure on purge gas. (Conditions: $T = 250\text{ }^{\circ}\text{C}$, $\text{H}_2/\text{CO}_2 = 3$, $\text{GHSV} = 500\text{ h}^{-1}$, and $R_p = 1\%$).

p	ΔF_{CO}	S3				S4		
		CO	CO_2	H_2	CH_3OH	CO	CO_2	CH_3OH
bar	kmol/h	kmol/h	kmol/h	kmol/h	kmol/h	kmol/h	kmol/h	kmol/h
10	125.9400	125.3935	101.2215	556.6007	8.2131	0.0040	0.2772	155.0586
20	14.4042	14.4434	34.1921	138.6663	1.2558	0.0162	2.8133	338.5694
30	5.6027	5.6083	21.0208	86.7610	0.5381	0.0172	4.5863	359.4812
40	2.9666	2.9759	14.5964	66.3010	0.3130	0.0169	5.8057	367.4122
50	1.8557	1.8434	10.8642	55.1952	0.2111	0.0164	6.6481	371.5195
60	1.2640	1.2455	8.3483	48.5353	0.1558	0.0156	7.1003	374.1384
70	0.9112	0.9037	6.7249	43.6764	0.1217	0.0149	7.4737	375.8047
80	0.6968	0.6819	5.4759	40.4375	0.0993	0.0141	7.5667	377.1396
90	0.5516	0.5362	4.6107	37.7053	0.0834	0.0135	7.6914	378.0403
100	0.4456	0.4347	3.9725	35.4008	0.0716	0.0131	7.8186	378.7310

Table 5Effect of H_2/CO_2 on purge gas. (Conditions: $T = 250\text{ }^{\circ}\text{C}$, $p = 50\text{ bar}$, $\text{GHSV} = 500\text{ h}^{-1}$, and $R_p = 1\%$).

H_2/CO_2	ΔF_{CO}	S3				S4		
		CO	CO_2	H_2	CH_3OH	CO	CO_2	CH_3OH
/	kmol/h	kmol/h	kmol/h	kmol/h	kmol/h	kmol/h	kmol/h	kmol/h
2.0	17.7646	17.6570	113.3554	36.9786	0.7730	0.0470	17.0885	241.6004
2.2	12.8877	12.9680	89.5398	36.8851	0.6255	0.0470	18.5646	269.5012
2.4	8.8898	8.8816	65.7099	36.8223	0.4796	0.0441	19.0562	297.0668
2.6	5.6365	5.5876	43.3373	37.4532	0.3473	0.0380	17.8331	324.1222
2.8	3.2058	3.2089	24.1968	40.9147	0.2459	0.0283	13.6810	349.8919
3.0	1.8564	1.8450	10.8818	55.1521	0.2111	0.0164	6.6607	371.5042
3.2	1.3552	1.3409	4.5481	99.3244	0.2983	0.0077	1.8755	382.9631
3.4	1.2511	1.2452	2.5558	167.1391	0.4731	0.0044	0.6565	386.0879
3.6	1.2508	1.2464	1.7897	242.0388	0.6736	0.0031	0.3217	386.9805
3.8	1.2784	1.2749	1.4038	319.2977	0.8824	0.0024	0.1922	387.2565
4.0	1.3093	1.3137	1.1769	396.6846	1.0923	0.0020	0.1300	387.2981

Table 6Effect of GHSV on purge gas. (Conditions: $T = 250\text{ }^{\circ}\text{C}$, $p = 50\text{ bar}$, $\text{H}_2/\text{CO}_2 = 3$, and $R_p = 1\%$).

GHSV	ΔF_{CO}	S3				S4		
		CO	CO_2	H_2	CH_3OH	CO	CO_2	CH_3OH
h^{-1}	kmol/h	kmol/h	kmol/h	kmol/h	kmol/h	kmol/h	kmol/h	kmol/h
100	1.8028	1.7847	10.3068	56.1474	0.1879	0.0118	7.4233	371.3487
500	1.7873	1.7847	10.3072	56.1446	0.1879	0.0118	7.4239	371.3447
1000	1.8036	1.7844	10.3045	56.1548	0.1879	0.0118	7.4208	371.3477
2000	1.7874	1.7844	10.3039	56.1572	0.1879	0.0118	7.4207	371.3582
3000	1.8041	1.7851	10.3163	56.1452	0.1879	0.0118	7.4288	371.3275
4000	1.8027	1.7835	10.3141	56.2464	0.1882	0.0118	7.4169	371.3321
5000	1.7929	1.7822	10.3433	56.4241	0.1887	0.0118	7.4157	371.3201
6000	1.8112	1.7811	10.3973	56.6607	0.1895	0.0117	7.4212	371.2401
7000	1.7838	1.7808	10.4846	56.9553	0.1906	0.0116	7.4409	371.1652
8000	1.7947	1.7757	10.5442	57.4448	0.1921	0.0115	7.4217	370.9954
9000	1.7784	1.7732	10.6421	57.9061	0.1936	0.0114	7.4291	370.8640
10000	1.7784	1.7692	10.7349	58.4841	0.1954	0.0113	7.4223	370.7834
20000	1.7773	1.7610	12.3610	64.2256	0.2165	0.0101	7.7042	368.8627

3.2. Effect of process parameters on purge gas

In the reaction system of CO_2 hydrogenation to methanol expressed in Eqs. (1)–(3), when CO_2 and H_2 are used as raw materials, the reaction products are CO and CH_3OH , and the CH_3OH selectivity is reduced owing to the by-product CO generation. When a recycle process as shown in Fig. 1 is used, the generated CO by the reaction is recycled to the reactor. According to the reaction Eqs. (2) and (3), CO may be a reaction product or a reactant, and the difference value between the molar flow rate of CO in the inlet stream (S5) and outlet stream (S6) in the reactor (as shown in Fig. 1) is defined as:

$$\Delta F_{\text{CO}} = F_{\text{CO},\text{S6}} - F_{\text{CO},\text{S5}} \quad (4)$$

The ΔF_{CO} represents the net production of CO within the boundary shown in Fig. 1. If $\Delta F_{\text{CO}} < 0$, CO is a reactant, if $\Delta F_{\text{CO}} > 0$, CO is a reaction product. The net production of CO ΔF_{CO} must be removed from the system through purge gas, otherwise, it will result in the accumulation of CO in the reaction system.

In this study, the calculation is based on a CO_2 feed rate of 391.01 kmol/h and process parameters in the range of $T = 200\text{--}300\text{ }^{\circ}\text{C}$, $p = 10\text{--}100\text{ bar}$, $\text{H}_2/\text{CO}_2 = 2\text{--}4$, $\text{GHSV} = 100\text{--}20000\text{ h}^{-1}$, and $R_p = 1\%$. The effect of process parameters on purge gas are shown in Tables 3–6.

As shown in Table 3, with the increment of reaction temperature, the amount of net CO generated ΔF_{CO} in the reaction system continues to increase. The flow rate of CO in stream S4 is relatively low owing to the small CO solubility in the flash liquid phase. To avoid the CO

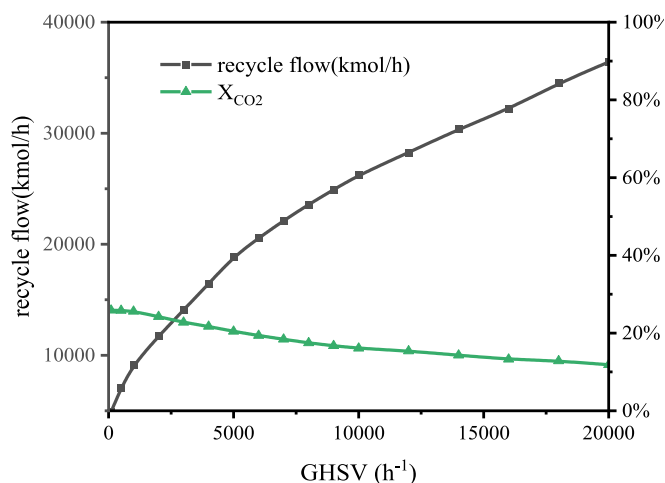


Fig. 6. Effect of GHSV on CO₂ conversion and recycling. (Conditions: $T = 250\text{ }^{\circ}\text{C}$, $p = 250\text{ bar}$, $\text{H}_2/\text{CO}_2 = 3$, and $R_p = 1\%$).

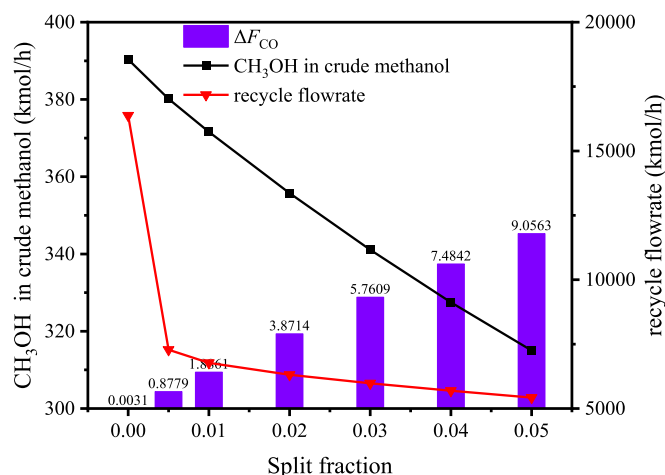


Fig. 7. Effects of the split fraction on the net production of CO, CH₃OH yield and recycle flowrate. (Conditions: $T = 250\text{ }^{\circ}\text{C}$, $p = 50\text{ bar}$, $\text{H}_2/\text{CO}_2 = 3$, and $\text{GHSV} = 500\text{ h}^{-1}$).

accumulation in the reaction system, ΔF_{CO} is mainly discharged via stream S3, it is accompanied by the emission of CO₂ and H₂ which is much higher than that of CO. Within the boundary shown in Fig. 1, based on the reaction stoichiometric ratio feed conditions ($\text{H}_2/\text{CO}_2 = 3$), if CO₂ and CO are discharged from the system, it is accompanied by the discharge of excess H₂. The amount of H₂ is approximately 3 times the sum of CO₂ and 2 times the CO emitted from the system. For example, at $300\text{ }^{\circ}\text{C}$, the H₂ loss of 100.6119 kmol/h accounts for 8.58% of the 1173.03 kmol/h H₂ feed in the whole system. Therefore, although the net amount of generated CO by the system is small, the resulting CO₂ and H₂ emissions are not ignored. In particular, the loss of H₂ with high value will increase the operating cost of the whole system.

As shown in Table 4, ΔF_{CO} in the reaction system decreases with the growth of the reaction pressure, especially when the pressure is higher than 50 bar, ΔF_{CO} is less than 2 kmol/h. Moreover, the reduction of the purge gas tends to decline slowly with the increase of the pressure, especially if the pressure is above 50 bar. Consistent with the effect of temperature on the purge gas, the net CO production of CO is small, but the resulting CO₂ and H₂ emissions, especially the H₂ loss, cannot be ignored. However, increasing the pressure will bring about the increment in energy consumption and equipment investment.

As can be seen from Table 5, ΔF_{CO} in the reaction system decreases as

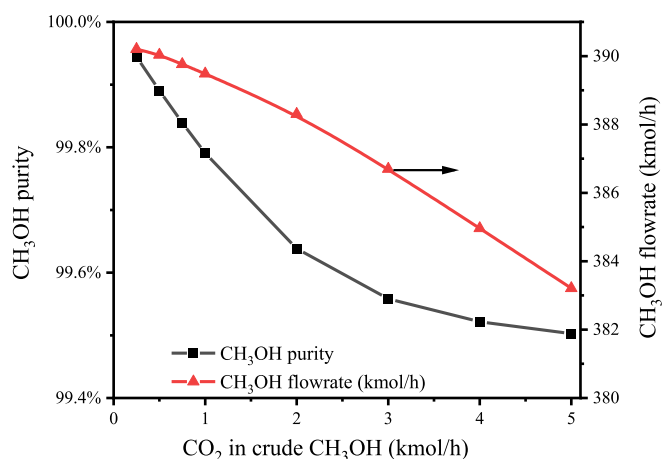


Fig. 8. Effect of CO₂ solubility on the CH₃OH purity and CH₃OH flowrate.

H₂/CO₂ grows, especially ΔF_{CO} tends to remain constant at $\text{H}_2/\text{CO}_2 > 3$. The effect of H₂/CO₂ on the amount of purge gas is different from the effect of temperature and pressure. At $\text{H}_2/\text{CO}_2 \leq 3$, the amount of purge gas depends on ΔF_{CO} that is similar to the effect of temperature and pressure, while at $\text{H}_2/\text{CO}_2 > 3$, the amount of purge gas depends on the excess H₂. For instance, when $\text{H}_2/\text{CO}_2 = 4$, the amount of H₂ loss is 396.6846 kmol/h , accounting for 25.36% of the feed H₂ amount of 1564.04 kmol/h . It is necessary to choose the appropriate H₂/CO₂, and if we want to achieve zero carbon emission, H₂/CO₂ should be set to 3.

As shown in Table 6, ΔF_{CO} remains almost constant in the range of GHSV from 100 to 20000 h^{-1} , while the amount of CO₂ and H₂ released in stream S3 grows, resulting in a declining CH₃OH yield in stream S4. The reason is that the one-way conversion rate of CO₂ reduces with the increment of GHSV and the amount of the recycle gas rises which leads to the loss of CO₂ and H₂ when R_p is set to 1%. To maximize CO₂ conversion and CH₃OH yield and reduce the recycling flow rate, it is advantageous to reduce GHSV. In this study, it is sufficient to choose GHSV less than 7000 h^{-1} .

Another key factor affecting the amount of purge gas is the split fraction (R_p , the ratio of purge gas flowrate to the vapor flowrate from the flash evaporator), as shown in Fig. 7. As can be seen from Fig. 7, with the reduction of R_p , ΔF_{CO} gradually lessens, and ΔF_{CO} is 0.0030 kmol/h when R_p is set to zero, which means that the net amount of generated CO in the whole reaction system is close to zero. The problem of purge gas resulting from the accumulation of CO can be circumvented. However, the recycle gas rapidly increases from 5425.53 kmol/h to 16377.72 kmol/h with the reduction of R_p ranging from 0.05 to 0, corresponding to that, the CO₂ to methanol carbon efficiency grows from 80.57% to 99.90%. Although the equipment size and investment may rise, it is still beneficial from increasing utilization efficiency of CO₂ and H₂.

Catalyst A exhibited superior performance compared to catalyst B in terms of CO₂ conversion and CH₃OH selectivity. To achieve a balanced relationship between leaching amount and methanol yield, selective optimization parameters were determined, which included reaction temperature, pressure, H₂/CO₂ feed ratio, and GHSV, set at $250\text{ }^{\circ}\text{C}$, 50 bar, 3, and less than 7000 h^{-1} , respectively. Despite variations in time, temperature, pressure, H₂/CO₂, or GHSV within the scope of process parameters, it is challenging to attain this target. Consequently, the sole resourceful approach is to attain zero net CO generation by the reaction system through zero purge gas.

3.3. Effect of CO₂ solubility in crude methanol

In addition to the loss of CO₂ in the purge gas, the crude CH₃OH contains a small amount of CO₂ that affects the CH₃OH purity and CH₃OH yield. As shown in Fig. 8, the CH₃OH yield is reduced with the

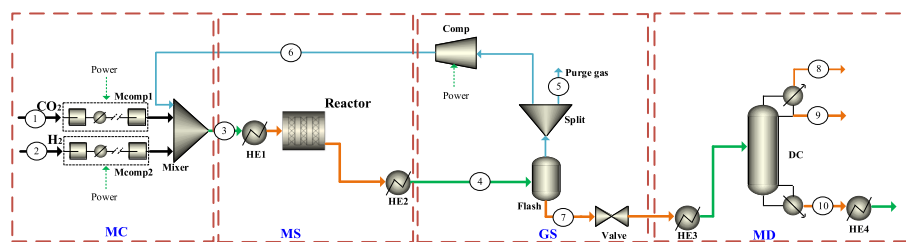


Fig. 9. Total recycling process after flash separation. (Process 1).

Table 7

Mass and energy balance of total recycling process after flash separation.

	S1	S2	S3	S4	S6	S7	S8	S9	S10
Phase	Vapor	Vapor	Vapor	/	Vapor	Liquid	Vapor	Liquid	Liquid
Temperature, °C	50.00	35.00	51.55	30.00	42.93	30.00	39.00	39.00	108.60
Pressure, bar	1.00	1.10	50.00	45.00	50.00	45.00	1.10	1.10	1.30
Vapor Fraction	1.00	1.00	1.00	0.95	1.00	0.00	1.00	0.00	0.00
Liquid Fraction	0.00	0.00	0.00	0.05	0.00	1.00	0.00	1.00	1.00
Enthalpy Flow, kW	-42641.76	94.09	-380570.93	-376440.08	-345790.05	-57220.95	-76.03	-26074.07	-30477.69
Mole Flows, kmol/hr	391.01	1173.03	17928.79	17149.89	16369.84	781.68	1.07	389.93	390.68
Mole Fractions									
CO	0.0000	0.0000	0.0021	0.0023	0.0024	0.0000	0.0000	0.0000	0.0000
CO ₂	1.0000	0.0000	0.1937	0.1611	0.1918	0.0006	0.5701	0.0002	0.0000
H ₂	0.0000	1.0000	0.8010	0.7773	0.8022	0.0004	0.2710	0.0000	0.0000
H ₂ O	0.0000	0.0000	0.0005	0.0285	0.0006	0.4995	0.0000	0.0000	0.9992
CH ₃ OH	0.0000	0.0000	0.0027	0.0308	0.0030	0.4995	0.1589	0.9998	0.0008
Mass Flows, kg/hr	17208.27	2364.69	184569.50	175300.69	167488.93	19570.75	32.88	12494.54	7043.03
Mass Fractions									
CO	0.0000	0.0000	0.0057	0.0063	0.0066	0.0000	0.0000	0.0000	0.0000
CO ₂	1.0000	0.0000	0.8282	0.6936	0.8249	0.0011	0.8165	0.0003	0.0000
H ₂	0.0000	1.0000	0.1568	0.1533	0.1581	0.0000	0.0178	0.0000	0.0000
H ₂ O	0.0000	0.0000	0.0009	0.0502	0.0011	0.3595	0.0000	0.0000	0.9985
CH ₃ OH	0.0000	0.0000	0.0084	0.0965	0.0094	0.6394	0.5903	0.9997	0.0015

increment of CO₂ in crude CH₃OH, and the CH₃OH purity also decreased gradually. Furthermore, the CH₃OH purity is unable to reach the industry requirement only through a single distillation column. It is requisite to control the content of CO₂ flowing into the distillation column and study CO₂ recovery in the liquid phase.

4. Process modification

Based on the process analysis for reaction unit with recycling, even if the generated CO in the system is small amount, the loss of CO₂ and H₂ cannot be ignored. The key to reduce the CO₂ and H₂ losses is to lessen the difference between inlet and outlet CO flow rate (ΔF_{CO}). Another, the CO₂ recovery from crude methanol also is an important issue. To maximize the utilization of CO₂ and H₂, two near-zero carbon emission processes with the total recycling are modified based on process in Fig. 1. The one is named total recycling process after flash separation and the other is named total recycling process after stripping separation. The Cu/Zn/Al/Zr fiber catalyst (Catalyst A) with higher activity is used to design a plant capacity of 100 kt/year of methanol in this study. The

selected PEM hydrogen production technology can adapt to the intermittency and fluctuation of renewable energy generation. The hydrogen (99.9 vol%) is fed into the methanol synthesis unit through the gas treatment system. The feed H₂ is gradually pressurized to 50 bar using a multistage compressor with an inter-cooler. The reactor for CO₂ hydrogenation to methanol is a multi-tube catalytic reactor with a tube length of 12 m, tube diameter of 0.06 m, and a catalyst bed voidage of 0.5.

4.1. Process 1: total recycling process after flash separation

Fig. 9 presents the total recycling process after flash separation for CO₂ hydrogenation to CH₃OH. The large amounts of residual CO₂, CO, and H₂ in the reaction products are separated by a flash evaporator (Flash 1), and then are recycled back to the reactor. To realize the near-zero carbon emission, the split fraction R_p for purge gas is set to zero. The liquid phase from the flash evaporator is purified through a single distillation column. CH₃OH and non-condensable CO₂ are separated from the top of the distillation column. The mass and energy balances of

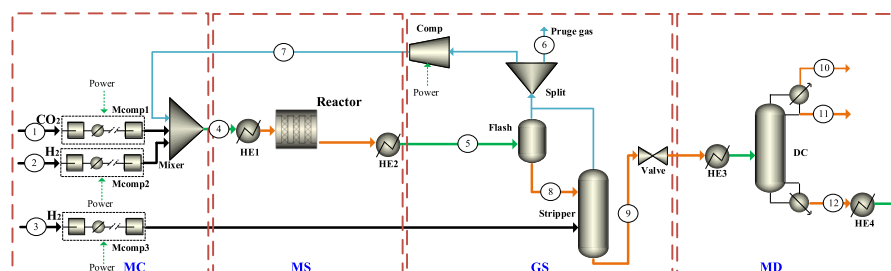


Fig. 10. Total recycling process after stripping separation. (Process 2).

Table 8

Mass and energy balance of total recycling process after stripping separation.

	S1	S2	S3	S4	S5	S7	S8	S9	S10	S11	S12
Phase	Vapor	Vapor	Vapor	Vapor	/	Vapor	Liquid	Liquid	Vapor	Liquid	Liquid
Temperature, °C	20.00	35.00	35.00	52.03	30.00	42.22	30.00	57.94	39.00	39.00	107.04
Pressure, bar	1.00	1.10	1.10	50.00	45.00	50.00	45.00	45.00	1.10	1.10	1.30
Vapor Fraction	1.00	1.00	1.00	1.00	0.90	1.00	0.00	0.00	1.00	0.00	0.00
Liquid Fraction	0.00	0.00	0.00	0.00	0.10	0.00	1.00	1.00	0.00	1.00	1.00
Enthalpy Flow, kW	-42765.30	33.20	61.09	-192100	-208559	-150158	-58632.01	-56844.24	-8.56	-25728.90	-30356.90
Mole Flows, kmol/hr	391.01	413.00	760.00	8605.46	7823.62	7801.45	793.40	782.24	0.50	390.66	391.08
Mole Fractions											
CO	0.0000	0.0000	0.0000	0.0244	0.0268	0.0269	0.0000	0.0000	0.0000	0.0000	0.0000
CO ₂	1.0000	0.0000	0.0000	0.1975	0.1673	0.1678	0.0111	0.0000	0.0000	0.0000	0.0000
H ₂	0.0000	1.0000	1.0000	0.7748	0.7024	0.8017	0.0002	0.0005	0.6609	0.0001	0.0000
H ₂ O	0.0000	0.0000	0.0000	0.0005	0.0505	0.0006	0.4932	0.4998	0.0000	0.0002	0.9994
CH ₄ O	0.0000	0.0000	0.0000	0.0027	0.0530	0.0030	0.4954	0.4997	0.3391	0.9997	0.0006
Mass Flows, kg/hr	17208.28	832.56	1532.07	94981.30	94981.30	76940.46	20033.32	19569.26	5.65	12515.08	7048.53
Mass Fractions											
CO	0.0000	0.0000	0.0000	0.0619	0.0619	0.0764	0.0000	0.0000	0.0000	0.0000	0.0000
CO ₂	1.0000	1.0000	0.0000	0.7878	0.6066	0.7488	0.0194	0.0000	0.0002	0.0000	0.0000
H ₂	0.0000	0.0000	1.0000	0.1415	0.1166	0.1639	0.0000	0.0000	0.1092	0.0000	0.0000
H ₂ O	0.0000	0.0000	0.0000	0.0008	0.0750	0.0010	0.3519	0.3599	0.0000	0.0001	0.9990
CH ₄ O	0.0000	0.0000	0.0000	0.0080	0.1399	0.0098	0.6286	0.6401	0.8906	0.9999	0.0010

the Process 1 are shown in Table 7.

4.2. Process 2: total recycling process after stripping separation

An additional stripper is introduced to modify the recycling process, as shown in Fig. 10. A portion of the feed H₂ is used to separate the dissolved CO and CO₂ from the liquid stream of the flash. Then the stripping gas phase is mixed with the gas phase from Flash unit as recycling gas. The liquid stream of the flash and the feed H₂ flow in counter-current mode in the stripping column. The mass and energy balance of the process 2 is shown in Table 8.

4.3. Utilization efficiency of CO₂ and H₂

The utilization efficiency of CO₂ and H₂ are the key indicators of CO₂ hydrogenation to CH₃OH, especially H₂ utilization has an important influence on the production cost (Luu et al., 2016). In this study, the utilization efficiency of CO₂ and H₂ refer to the conversion ratio of feed composition to CH₃OH, as shown in Eqs. (5) and (6).

$$E_{\text{CO}_2} = \frac{F_{\text{CH}_3\text{OH,product}}}{F_{\text{CO}_2,\text{Feed}}} \times 100\% \quad (5)$$

$$E_{\text{H}_2} = \frac{2 \times F_{\text{CH}_3\text{OH,product}}}{F_{\text{H}_2,\text{Feed}}} \times 100\% \quad (6)$$

According to the results in Table 7, the E_{CO_2} and E_{H_2} for process 1 are 99.70%, 66.47%, respectively. The loss of carbon and hydrogen is mainly in the form of CO₂, H₂, and CH₃OH, which is discharged from the top of the distillation column. Similarly, E_{CO_2} and E_{H_2} for process 2 are respectively 99.88%, 66.59% according to Table 8. In the CO₂ hydrogenation to CH₃OH reaction, 1 mol of H₂ must be consumed to produce 1 mol of water, that is, the maximum theoretical the utilization efficiency of H₂ of this reaction is 66.67%. Therefore, the utilization efficiency of H₂ for processes 1 and 2 reaches a relatively high level. Compared to the current typical results in Table 2, both two processes have good feedstock utilization and atomic economy, that is why it is called the near-zero carbon emission processes.

5. Process integration

To explore the energy utilization of the process, the integration of the process is analyzed according to the process simulation results.

5.1. Process energy integration

Pinch analysis is used to minimize the energy consumption of the above two processes. The detailed data of hot and cold streams, including initial temperature, target temperature, heat capacity flow rate, and heat load, are extracted respectively, referring to supporting information (Tables S5–S7). In particular, the initial heat exchange

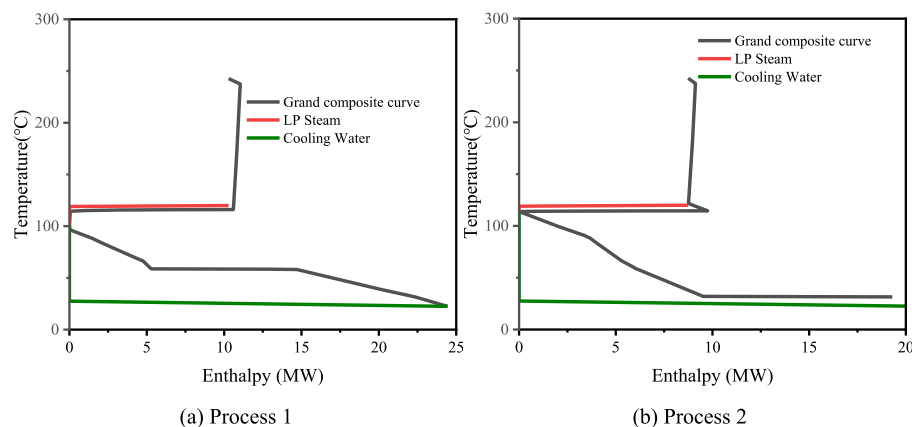


Fig. 11. Grand composite curve for two near-zero carbon emission processes.

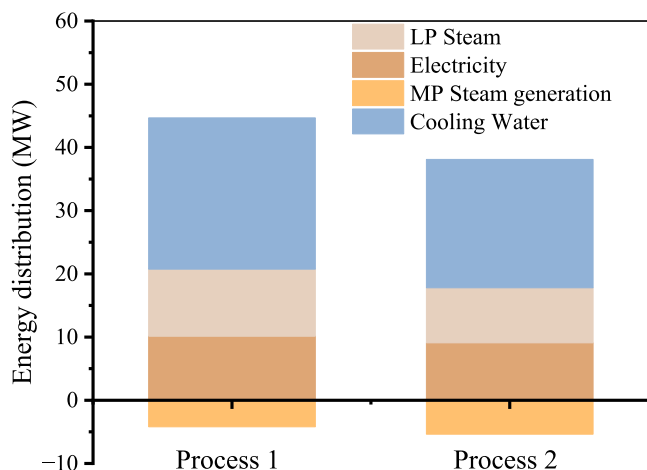


Fig. 12. Distribution of energy consumption.

network is not included the exothermic heat of the reactor, and the removal heat unit of the reactor is treated separately. In the heat exchange network, the minimum temperature difference ($\Delta T_{\min}$) is assumed 15 °C. The grand composite curve of the two processes are presented in Fig. 11. For process 1, the hot utility is 10.70 MW of low-pressure steam (LP steam) and the cold utility is 24.44 MW of cooling water. For Process 2, the hot and cold utilities are 8.75 MW of LP steam and 20.34 MW of cooling water, respectively. The designed heat exchanger network is shown in supporting information (Fig. S1).

The power consumption of compressors that compress CO_2 , H_2 , and recycle gas is also an important part of energy consumption. Combined with the power consumption and reaction heat generation for the two processes, the distribution of the energy consumption of the two processes is shown in Fig. 12. It shows that the total recycling process after stripping separation with the highly active Cu/Zn/Al/Zr fiber catalyst has an obvious energy-saving effect.

The generated reaction heat from the reactor is continuously removed by generating medium-pressure steam (MP Steam Generation). Loads of MP Steam are respectively 4.20 MW and 5.37 MW for Process 1 and Process 2, which can further reduce the amount of the LP steam. Once some of the LP steam is offset and lowered, it still requires 6.50 MW and 3.38 MW LP steam for Process 1 and Process 2, respectively. However, for Process 2, the heat load of MP steam produced by the

reactor is more than half of the LP steam requirement of the reboiler in the crude CH_3OH distillation column. If double-effect distillation is used, the modified process will not require external heating utility but only with the need of cold utility.

5.2. Process integration with double-effect distillation

Multi-effect distillation is established for the syngas conversion to CH_3OH to improve the utilization of the steam at the top of the column in the distillation process (Li et al., 2022; Zhu et al., 2021). In the industrial syngas to CH_3OH processes, the CH_3OH mass fraction of crude CH_3OH is about 80 %–95%, while the CH_3OH mass fraction of crude CH_3OH for CO_2 hydrogenation to CH_3OH is about 60% because 1 mol of CH_3OH is formed accompanied by 1 mol of water. According to the different feed methods and the gradient direction of the column pressure, there are three kinds of typical double-effect distillation configurations are obtained to achieve energy efficient target, called the feed splitting (FS), the light split/forward (LSF), and the light split/reverse (LSR) as shown in Fig. 13.

In the light split/forward (LSF), the raw materials all enter the high-pressure tower, the top of the tower to get part of methanol production, the bottom solution is fed into the atmospheric tower, the top of the atmospheric tower to get part of methanol production, the bottom of the tower discharge waste water. The light split/reverse (LSR), raw materials first all into the atmospheric tower, the top of the tower to get part of methanol production, the bottom of the tower solution into the high-pressure tower, the top of the high-pressure tower to get part of methanol production, the bottom of the tower discharge wastewater. In the feed splitting (FS), the concentration of raw materials entering the high-pressure column and the atmospheric column is the same, methanol products are obtained from the top of the two towers, and waste water is discharged from the bottom of the two towers.

In this study, the top pressure of the low-pressure column (LP column) is set to be atmospheric. The top pressure of the high-pressure column (HP column) is varied so that the temperature difference between the condenser temperature of the HP column and the reboiler temperature of the LP column meets the requirements of $\Delta T_{\min}$. The total number of stages and feed locations can be designed as variables to minimize total annual cost for further investigation. The reflux ratio and distillate rate are varied so that the heat transfer load is matched between the condenser duty of the HP column and the reboiler duty of the LP column and the purity of CH_3OH products meets the requirements. The operating parameters and results of the three typical double-effect

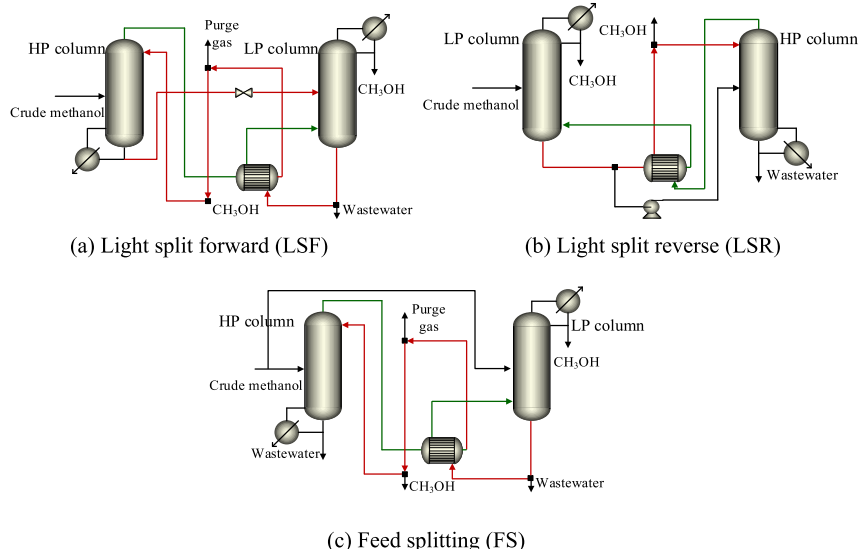
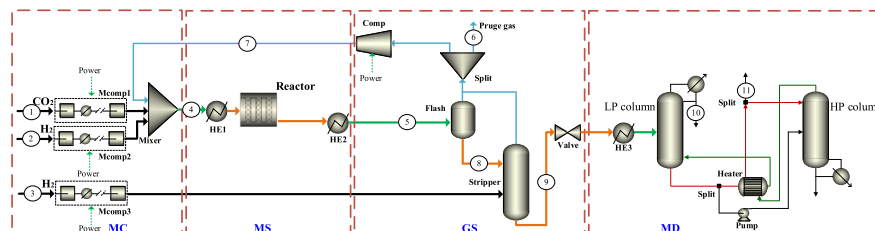
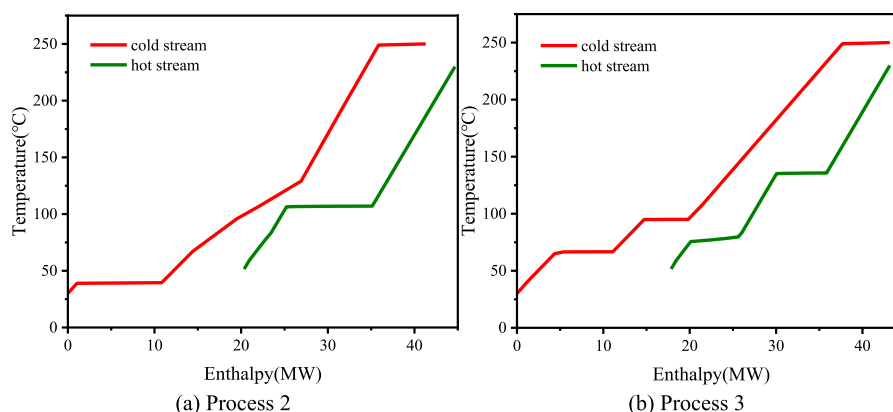


Fig. 13. Three typical double-effect distillation configurations.

Table 9

Parameters and results of three typical double-effect distillation.

Items	Distillation with LSF		Distillation with LSR		Distillation with FS	
	LP Column	HP Column	LP Column	HP Column	LP Column	HP Column
Feedstock, kg/hr	16085.60	19569.42	19569.42	13111.61	10386.21	9183.21
Overhead product, kg/hr	9038.43	3483.82	6457.81	6065.54	6645.21	5878.00
Mass reflux ratio	1.54	5.71	2.22	1.92	2.12	2.65
Number of stages	30	40	40	30	40	30
Feed stage	20	15	15	20	15	20
Condenser temperature, °C	66.64	122.00	65.00	94.97	65.00	117.00
Bottom temperature, °C	107.05	135.18	79.66	135.69	102.23	160.07
Condenser pressure, bar	1.10	7.00	1.10	3.00	1.10	7.00
Condenser duty, kW	−6994.89	−6174.55	−6319.59	−5076.76	−6327.23	−5758.35
Reboiler duty, kW	6174.54	6244.83	5076.31	5751.76	5759.29	5856.90

**Fig. 14.** Improved process 2 with double-effect distillation with LSR. (process 3).**Fig. 15.** Composite curve for process 2 and process 3.

distillation processes are shown in Table 9. In terms of column pressure, the pressures of the LP column and the HP column for LSR are respectively 1.1 bar and 3.0 bar, which are lower than that of other two configurations. The reboiler duty of the HP column represents the steam demand for double-effect distillation. This reboiler duty for LSR is 5751.76 kW and is the least among the three typical configurations. Therefore, both in terms of energy utilization and equipment investment, the double-effect distillation with LSR may be the suitable configuration for the separation of crude methanol from CO₂ hydrogenation.

5.3. Process 3: improved process 2 with double-effect distillation with LSR

The double-effect distillation of LSR is used to improve the process 2, this integrated process is denoted as process 3 (shown in Fig. 14), which is expected to further reduce the external heating utilities.

The composite curve of process 3 is shown in Fig. 15, which has no thermal utility and a load of cooling water of 13.70 MW. Furthermore, process 3 also generates excess energies for 0.42 MW of LP steam and

3.73 MW of MP steam respectively. Mass, energy balance and the designed heat exchanger network of process 3 are shown in supporting information (Table S11 and Fig. S2).

6. Techno-economic analysis

The key parameters of each unit of the processes are shown in Table S8-Table S10.

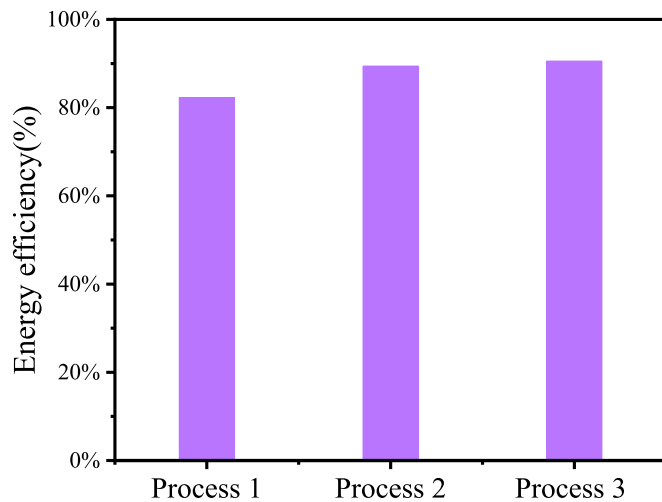
6.1. Technical analysis

The key parameters and consumption results for the three kinds of total recycling processes are exhibited in Table 10. The overall CH₃OH yield reaches 99.88%, that is the near-zero CO₂ emissions have been achieved in terms of the utilization of feed CO₂. The heat duty of the reboiler of process 3 using double-effect distillation with LSF is the least of the three configurations. The steam usage of process 3 is −0.60 ton steam/ton CH₃OH, which means the steam produced is excess as a by-product in process 3. Thus, the total recycling process combined

Table 10Key performance indicators and consumption results for CO₂ hydrogenation to CH₃OH.

Parameter	Unit	Process 1	Process 2	Process 3	Kiss et al. (2016)	Dif. 1	Dif. 2	Dif.3
CH ₃ OH production rate	kton/year	99.94	100.12	100.12	100.07	−0.13%	0.05%	0.05%
Purge to feed ratio	mol/mol	0	0	0	0	/	/	/
Recycle to feed ratio	mol/mol	10.46	4.99	4.99	5.47	91.22%	−8.78%	−8.78%
H ₂ : CO ₂ ratio (feed/reactor inlet)	mol/mol	3/4.13	3/3.92	3/3.92	3.0/2.84	/	/	/
H ₂ conversion in reactor (per pass)	%	7.18	17.59	17.59	18.17	−60.48%	−3.19%	−3.19%
CO ₂ conversion (per pass)	%	20.44	22.99	22.99	17.20	18.84%	33.66%	33.66%
Utilization efficiency of CO ₂	%	99.70	99.88	99.89	99.83	−0.13%	0.05%	0.06%
Utilization efficiency of H ₂	%	66.47	66.59	66.59	66.55	−0.12%	0.06%	0.06%
CH ₃ OH yield (overall process)	%	99.70	99.88	99.89	99.83	−0.13%	0.05%	0.06%
Power of H ₂ feed compressor (MCOMP1)	kW	6319.91	6241.20	6241.20	5962.96	5.99%	4.67%	4.67%
Power of recycle compressor (MCOMP2)	kW	1739.84	793.04	793.04	911.96	90.78%	−13.04%	−13.04%
Heat generated in the reactor	kW	−4951.75	−5759.70	−5759.70	−4448.72	11.31%	29.47%	29.47%
Heat duty of reboiler (distillation column)	kW	10531.92	9839.48	5751.76	7618.42	38.24%	29.15%	−24.50%
Electricity usage (per ton methanol)	kWh/ton CH ₃ OH	810.57	725.33	725.33	550 ^a	47.38%	31.88%	31.88%
Steam usage (per ton methanol)	ton steam/ton CH ₃ OH	1.41	1.15	−0.60	1.16	21.55%	−0.86%	−151.72%
Electricity cost (at 0.08 €/kWh)	Euro/ton CH ₃ OH	64.84	58.02	58.02	49.37	31.33%	17.52%	17.52%
Steam cost (at 25 €/ton steam)	Euro/ton CH ₃ OH	35.16	28.67	−14.94	29.03	21.12%	−1.24	−151.46
Pure CO ₂ use (per unit of CH ₃ OH product)	kg/kg	1.3792	1.3767	1.3767	1.3758	0.25%	0.07%	0.07%
Pure H ₂ use (per unit of CH ₃ OH product)	kg/kg	0.1881	0.1877	0.1877	0.1892	−0.58%	−0.79%	−0.79%

^a Note: The CO₂ feeding pressure in the reference is 100 bar, while the CO₂ of feed in this study is 1.1 bar. Dif. 1 is difference between process 1 and reference. Dif. 2 is difference between process 2 and reference. Dif. 3 is difference between process 3 and reference.

**Fig. 16.** Energy efficiency for processes.

stripping separation with double-effect distillation shows the advantages of higher resource efficiency and lower energy consumption.

In addition to the key parameters and consumption results, energy use efficiency is also an important technical index for evaluating the different processes of CO₂ hydrogenation to methanol. Energy efficiency

(η) refers to the ratio of the sum of product energy, raw material energy and process energy consumption, which is defined in Equation (7) (Zhang et al., 2020).

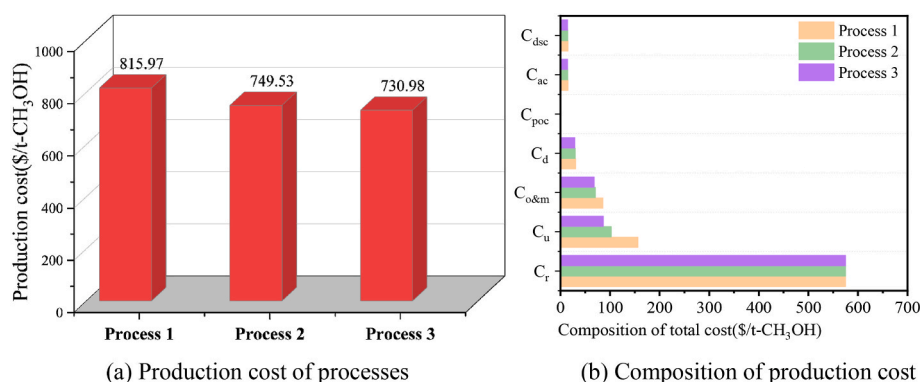
$$\eta = \frac{E_{\text{product}}}{E_{\text{feedstock}} + E_{\text{utilities}}} \quad (7)$$

Where, E_{product} is the energy of output product, $E_{\text{feedstock}}$ is the energy of input raw material, and $E_{\text{utilities}}$ is the consumption of utilities in the whole process.

The energy efficiency of CO₂ hydrogenation to methanol processes is shown in Fig. 16. The energy efficiency of the processes are 82.30%, 89.44%, and 90.54%, respectively. The input energy and output energy of CO₂ hydrogenation to methanol process in three different processes are counted. The input energy includes raw hydrogen and the consumption of utilities. In terms of input energy, H₂ input energy in different processes is basically the same, while that of utilities is quite different.

6.2. Economic analysis

The production cost (PC) is an important indicator to evaluate the economy of a process, including the fixed capital investment (FCI) and the working capital, which is the cost of purchasing equipment and installation costs, and the operating capital, which is needed to maintain the normal operation of the project. In this study, calculation formulas and data economic analysis are adopted in the literature (Zhou et al.,

**Fig. 17.** Production cost of processes.

2023; Wang et al., 2022; Dey et al., 2021; Mridha et al., 2023), which have been listed in the supporting information (Tables S13–S16). Furthermore, the calculation process is based on an annual operation time of 8000 h year⁻¹. The results of the economic analysis are presented in Fig. 17.

$$EI = \sum_j \theta \cdot EI_j^r \cdot \left(\frac{S_j}{S_j^r} \right)^{sf} \quad (8)$$

$$FCI = EI \cdot \left(1 + \sum_{i=1}^n RF_i \right) \quad (9)$$

$$PC = C_r + C_u + C_{o\&m} + C_d + C_{poc} + C_{ac} + C_{dsc} \quad (10)$$

Where θ represents the domestic production index, EI_j^r represents the reference of equipment j ; S_j represents the production scale of equipment j ; S_j^r represents the reference scale of equipment j ; sf refers to the scale index; RF_i is the scaling factor; C_r is the cost of raw materials; C_u is utility costs; $C_{o\&m}$ is the cost of maintenance and operation; C_d is the cost of folding; C_{poc} is factory overhead; C_{ac} is administrative expenses; C_{dsc} is selling expenses.

According to Eqs. 8–10, the production costs of processes are 815.97 \$/t-CH₃OH, 749.53\$/t-CH₃OH, and 730.98 \$/t-CH₃OH for each process. Compared with the process 1, the production cost reduces by 8.14% for process 2 and by 10.42% for process 3. Since the price of hydrogen using renewable energy is 3.13\$/kg, the raw material cost accounts for 83.07%, 76.73% and 78.68% respectively, which the cost of this process is relatively high. When the price of electricity is lowered to 0.1 CNY·(kW·h⁻¹), the production costs of processes are 347.80 \$/t-CH₃OH, 308.15 \$/t-CH₃OH, and 289.60 \$/t-CH₃OH for each process, which shows good economic value for CO₂ hydrogenation to CH₃OH.

7. Conclusion

In the CO₂ hydrogenation to methanol process, the purge gas produced by net CO generation in the reaction system leads to the loss of CO₂ and H₂, thereby reducing the efficiency of CO₂ and H₂ utilization. Achieving near-zero carbon emissions relies on optimizing the total recycling, which can be achieved through the use of highly selective catalysts and the addition of stripping to reduce the amount of cycle. The main conclusions are as follows.

1. The carbon utilization efficiency are greater than 99.70% for the proposed total recycling processes, which is deemed as near-zero carbon emission processes.
2. After process integration, the hot and cold utilities for process 1 are 10.70 MW of LP steam and 24.44 MW of cooling water, respectively.

Nomenclature

KIST	Korea Institute of Science and Technology
NIRE/RITE	National Institute for Resources and Environment
ALK	alkaline water electrolysis
PEM	proton exchange membrane electrolysis
SOEC	solid oxide electrolysis

T temperature

<i>p</i>	pressure
GHSV	gas firing hourly space velocity
H ₂ /CO ₂	hydrogen to carbon molar ratio
<i>S</i> _{CO}	selectivity of CO
<i>X</i> _{CO2}	conversion of CO ₂
<i>S</i> _{CH3OH}	selectivity of CH ₃ OH

Process 2 has 8.75 MW of LP steam and 20.34 MW of cooling water. After some of the hot steam is offset by the generated reaction heat and using double-effect distillation of LSR, the utility for process 3 is just 13.70 MW of cooling water.

3. The techno-economic evaluation shows that the production costs for the three processes are 815.97 \$/t-CH₃OH, 749.53\$/t-CH₃OH, and 730.98 \$/t-CH₃OH, respectively. When the price of electricity is lowered to 0.1 CNY·(kW·h⁻¹), the production costs of processes are 347.80 \$/t-CH₃OH, 308.15 \$/t-CH₃OH, and 289.60 \$/t-CH₃OH for each process, which shows good economic value for CO₂ hydrogenation to CH₃OH

This study promises to become one of the key pathways for decarbonizing sectors as industry in view of the energy transition towards net zero greenhouse gas emissions.

CRediT authorship contribution statement

Dongliang Wang: Conceptualization, Funding acquisition, Formal analysis. **Jingwei Li:** Methodology, Investigation, Data curation, Writing – original draft. **Wenliang Meng:** Formal analysis, Writing – review & editing. **Zuwei Liao:** Conceptualization, Software, Writing – original draft. **Siyu Yang:** Software, Investigation, Data curation. **Xiaodong Hong:** Conceptualization, Formal analysis. **Huairong Zhou:** Formal analysis, Data curation. **Yong Yang:** Software, Resources. **Guixian Li:** Methodology, Supervision, Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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R_p	ratio of purge gas to the vapor from the flash evaporator
ΔF_{CO}	net production of CO within the boundary
MC	gas compression
MS	material synthesis
GS	gas separation
MD	methanol distillation
COMP	compressor
MCOMP	multistage compressor
DC	Distillation column
ΔT_{min}	minimum heat transfer temperature difference
GCC	grand composite curve
E_{CO_2}	utilization efficiency of CO_2
$F_{CH_3OH,product}$	molar flowrate of CH_3OH in the product
$F_{CO_2,Feed}$	molar flowrate of CO_2 in the feed
E_{H_2}	utilization efficiency of H_2
$F_{H_2,Feed}$	molar flow rate of H_2 in the feed
LP steam	low-pressure steam
MP Steam Generation	generating medium-pressure steam
LSF	light split forward
LSR	light split reverse
FS	feed splitting
LP column	low-pressure column
HP column	high-pressure column
Dif	difference
η	Energy efficiency
$E_{product}$	the energy of output product
$E_{feedstock}$	the energy of input raw material
$E_{utilities}$	the consumption of utilities in the whole process
PC	production cost
FCI	fixed capital investment
θ	the domestic production index
El_j^r	the reference of equipment j
S_j	the production scale of equipment j
S_j^r	the reference scale of equipment j
sf	the scale index
RF_i	the scaling factor
C_r	the cost of raw materials
C_u	utility costs
$C_{o\&m}$	the cost of maintenance and operation
C_{poc}	factory overhead
C_{ac}	administrative expenses
C_{dsc}	selling expenses

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jclepro.2023.137388>.

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